

By scalar multiplication, $x_1 \begin{bmatrix} 0 \\ 4 \\ -1 \end{bmatrix} + x_2 \begin{bmatrix} 1 \\ 6 \\ 3 \end{bmatrix} + x_3 \begin{bmatrix} 5 \\ -1 \\ -8 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

In Exercises 11 and 12, determine if b is a linear combination of a_1, a_2 and a_3 .

11. $a_1 = \begin{bmatrix} 1 \\ -2 \\ 0 \end{bmatrix}, a_2 = \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}, a_3 = \begin{bmatrix} 5 \\ -6 \\ 8 \end{bmatrix}, b = \begin{bmatrix} 2 \\ -1 \\ 6 \end{bmatrix}$

Solution: We need to determine if there exists $x_1, x_2, x_3 \in \mathbb{R}$ such that $x_1 a_1 + x_2 a_2 + x_3 a_3 = b$, where $x_1, x_2, x_3 \in \mathbb{R}$. This is equivalent to finding whether the augmented matrix $[a_1 \ a_2 \ a_3 \ b]$ has a solution set.

$$\begin{bmatrix} 1 & 0 & 5 & 2 \\ -2 & 1 & -6 & -1 \\ 0 & 2 & 8 & 6 \end{bmatrix} \xrightarrow[\substack{\text{Row 2} = \text{Row 2} \\ + 2 \times \text{Row 1}}]{\text{Row 3} = \text{Row 3} - 2 \times \text{Row 2}} \begin{bmatrix} 1 & 0 & 5 & 2 \\ 0 & 1 & 4 & 3 \\ 0 & 2 & 8 & 6 \end{bmatrix} \xrightarrow{-2 \times \text{Row 2}} \begin{bmatrix} 1 & 0 & 5 & 2 \\ 0 & 1 & 4 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

~~Row 1 = Row 1~~ Verification that whether the final matrix is in echelon form

- The condition that all the zero rows are below all the non-zero rows is True, since here the first two rows are non-zero and the third row is zero.
- The leading entry of Row 1 = 1 in Column 1
The leading entry of Row 2 = 1 in Column 2
∴ The condition that the leading entry of a Row is in a column to the right of the leading entry of all rows above it is True.
- Row 2 Column 1 = Row 3 Column 1 = 0
Row 3 Column 2 = 0
∴ The condition that each pivot column has only 0's below the pivot position of the column is True. $\square$
- ∴ The matrix $\begin{bmatrix} 1 & 0 & 5 & 2 \\ 0 & 1 & 4 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix}$ is in echelon form. $\square$